

Week 7 Deliverable: Application Development

Team Ghanti Tininini

King's College Nepal, Westcliff University

DATA 200: Applied Statistical Analysis

Professor Regmi

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Overview

This week we built a locally running Python application that operationalizes the Star Power Index models developed in Weeks 5 and 6. The application allows a user to enter a production budget, cast popularity scores, genre, and release season, and receive an instant predicted revenue alongside a profitability classification. It is built using Streamlit for the interface, with the trained OLS and logistic regression models loaded from serialized pickle files.

Libraries Used

Table 1 lists the Python libraries used in the application and the role each plays.

Table 1

Python Libraries and Their Purpose

Library	Purpose
Streamlit	Web-based user interface running locally in the browser
statsmodels	OLS regression model, trained on the TMDB dataset and serialised with pickle
scikit-learn	Logistic regression for ROI profitability classification
pandas	Data handling and feature preparation for model input
numpy	Log transformations applied to user inputs before prediction
pickle	Loading the pre-trained OLS and logistic models at runtime

requests	Optional live cast popularity lookup from the TMDB API by actor name
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Application Structure

The application is organized into three files. `app.py` contains the Streamlit interface and handles all user inputs and output display. `predictor.py` contains the prediction logic, loading the serialized models and applying the same log transformations used during training. `tmdb_api.py` handles optional live actor name lookups from the TMDB API so a user can type an actor's name rather than entering a manual popularity score. The folder structure is as follows.

```
star_power_index/
|-- app.py           # Streamlit interface
|-- predictor.py     # Model loading and prediction logic
|-- tmdb_api.py      # Optional live TMDB actor lookup
|-- models/
|   |-- ols_model.pkl # Trained OLS regression model
|   |-- logit_model.pkl # Trained logistic regression model
|-- requirements.txt
```

How the Applications Works

The user opens a terminal, navigates to the project folder, and runs the command below. The application opens automatically in the default browser at <http://localhost:8501>.

```
streamlit run app.py
```

From the interface, the user enters a production budget using a slider, three actor names or manual TMDB popularity scores, a primary genre from a dropdown list, and a release season.

On clicking Predict, the application log-transforms the budget and cast popularity inputs, constructs the feature vector, and passes it to the OLS model to produce a predicted revenue figure with a 95% confidence interval. The same inputs are passed to the logistic model, which returns a profitability label (Profitable or At Risk) along with the estimated probability. A bar chart comparing the predicted revenue to genre averages is also displayed for context.

Installation and Running Instructions

The application requires Python 3.11 or above. Dependencies are listed in requirements.txt and can be installed with the command below.

```
pip install -r requirements.txt
```

Before running the app for the first time, the models must be trained and saved using the training script included in the repository. This only needs to be done once.

```
python train_models.py  
streamlit run app.py
```

Conclusion

The Star Power Index Calculator is a fully working, locally running Python application that puts the statistical models from Weeks 5 and 6 into a usable tool. A user with no statistical background can enter a proposed cast and budget and receive a revenue prediction and profitability estimate within seconds. The application will be demonstrated live during the Week 8 final presentation.

References

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